

Benjamin T. Torres

Master's Candidate

Department of Biological Sciences
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Education

- 2025–Pres. : **Master of Science, Biological Sciences**, *California State University, Stanislaus*.
2020–2024 : **Bachelor of Science, Ecology, Evolution & Organismal Biology**, *California State University, Monterey Bay*.

Publications & Manuscripts

- 2026 : **Torres, B.T.**; Alter, S.E., Adaptive convergence and haplotype sharing reveal depth-related evolution of a low-light optimized gene in rockfishes (*Sebastes*), *Journal of Evolutionary Biology*. (*In Review*)
2026 : **Torres, B.T.**; McFarlan, S.; Yu, A.S.; Arun, A., A review of phylogenetic relationships in Lampropeltis: is population structure sufficient for delimiting diverse clades? (*Serpentes: Colubridae*), *Journal of Evolutionary Systematics*. (*In Preparation*)

Research Experience

CSU Stanislaus

- Aug., 2025 – ***Aquatic gartersnake evolutionary ecology (MS Thesis)***.
Present : Testing for niche conservation and niche divergence in Western aquatic gartersnakes using ecological niche modeling, identifying signals of functional morphological adaptation to heterogeneous habitat using museum specimens.
Advisor : **Dr. Marina M. Gerson**, Associate Professor, Department of Biological Sciences, CSU Stanislaus
CSU Monterey Bay
May, 2024 – ***Lead Lab Technician***.
Aug., 2025 : Optimized 3RADseq protocol, mentored undergraduate researchers, utilized RADseq for Canary Rockfish population structure study, trained to use Illumina platforms, including Miseq, identified species of interest in Pajaro river with eDNA.
Aug., 2023 – ***Rockfish systematics and speciation (Sebastes)***.
Aug., 2025 : Developed a comprehensive phylogeny using mitochondrial and nuclear DNA, determined possible drivers of speciation using positively selected regions, tested for positive and balancing selection in different clades, found a novel “depth threshold”, at which independent lineages undergo positive selection, collected, prepared, and sampled fish through CCFRP from various locations.
Jul., 2024 – ***Spade-foot toad conservation genomics***.
Oct., 2024 : Analysed environmental DNA using Illumina sequencing to determine endangered spade-foot toad population presence in Beale Air Force Base.
Jan., 2024 – ***Introgression among incipient rockfish species***.
May, 2024 : Performed ddRADseq on rockfish tissue samples to quantify population structure and introgression between *Sebastes carnatus* and *Sebastes crysomelas*.
Advisor : **Dr. Liz Alter**, Associate Professor, Department of Biology & Chemistry, CSUMB ([Personal Web-page](#))

May, 2024 – ***Rattlesnake behavioral assays.***

Oct., 2024 Conducted behavioral assays of the Northern Pacific Rattlesnake, assessed flight initiation distances (FIDs) and alert distances (ADs), explored impacts of repeated human encounters on rattlesnake reactions, surveyed protected wildlife areas, learned how to identify local herpetofauna to species level

Advisor : **Dr. Jennifer Duggan**, Associate Professor, Department of Environmental Science, CSUMB

Jun., 2023 – ***CSUMB Ocean Science REU.***

Aug., 2023 Refined skills in science writing and communication, developed an original research question, learned how to work with others as a fluid team, presented results at CSUMB Summer Research Symposium.

Advisor : **Dr. Palacios**, Associate Professor, Department of Biology & Chemistry, CSUMB

Marine Life Studies

Aug., 2022 – ***Cetacean population surveys.***

May., 2025 Close observation and recording of cetacean behavior, photographing of cetaceans for logging and identification - including underwater videography, assisted in cetacean disentanglements including attaching telemetry buoys, completed educational infographics and illustrations on observed species.

Advisor : **Stephanie Marcos**, Operations Manager, Marine Life Studies ([Institution Web-page](#))

Conference Presentations

Oral Presentations

2026 **Benjamin Torres**, Adaptive convergence and haplotype sharing reveal depth-related evolution of a low-light optimized gene in rockfishes (Sebastes) (flash talk), Joint Meeting of Ichthyologists and Herpetologists (JMIH) (*Scheduled*).

2026 **Benjamin Torres**, Assessing how climatic factors drive garter snake population distributions, CSU Stanislaus Science Day (public event).

2024 **Benjamin Torres**, Drivers of diversification in rockfishes (Genus *Sebastes*), CSUMB Capstone Festival.

2023 **Benjamin Torres**, Drivers of diversification in rockfishes (Genus *Sebastes*), CSUMB Summer Research Symposium.

Poster Presentations

2026 **Benjamin Torres**, Integrating morphometrics and ecological niche models to reveal patterns in landscape-scale evolutionary ecology in Western aquatic garter snakes (*Thamnophis*), Joint Meeting of Ichthyologists and Herpetologists (JMIH) (*Scheduled*).

2026 **Benjamin Torres**, Assessing how climatic factors drive garter snake population distributions, CSU Stanislaus Science Day (public event).

2024 **Benjamin Torres**, Drivers of diversification in rockfishes (Genus *Sebastes*), CSUMB Capstone Festival.

2023 **Benjamin Torres**, An updated phylogeny and drivers of diversification in rockfishes (Genus *Sebastes*) , Western Society of Naturalists.

Certifications & Trainings

2023 **Level 1 Venomous Snake Handling**, Save the Snakes, Snake Conservation Center, Sacramento, CA

2023 **CCFRP** - MPA assessment protocol, (Fish handling, identification, tissue sampling, morphometrics, dissection, descension)

2023 **CITI** - Responsible conduct of research

2023 **CITI** - Animal Care and Use: Working with Fish in Research Settings

Fellowships & Awards

2025 –present **Graduate Teaching Fellowship** California State University, Stanislaus.

Computer skills

Software R, PAST, Genious Prime, MEGA, Galaxy, Advanced Java

Referees

Dr. Liz Alter

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Biology & Chemistry*
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Dr. Marina M. Gerson

*Associate Professor, Department of
Biological Sciences*
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Dr. Brian Halstead

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